

MOHAMED HANY REDA

Full-Stack Software Engineer · AI & Machine Learning

JavaScript · Python · Flutter · FastAPI · PostgreSQL · PyTorch · Docker | 6+ years · led teams up to 10

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PROFESSIONAL SUMMARY

Full-stack software engineer with six years building web, mobile and AI products end to end. My work spans deterministic simulation and digital twins, reinforcement-learning decision systems, explainable computer vision, and multi-tenant enterprise architecture — each delivered as a working product with a data layer, an interface, and automated tests. I connect business goals, product requirements and engineering execution, and my engineering background lets me understand technical dependencies, identify delivery risks early, and make project decisions with real technical context. Working proficiency in English and Russian alongside native Arabic supports collaboration across international teams.

AVAILABILITY

Available on-site, remotely or hybrid, and open to business trips. Looking for full-time employment.

CORE COMPETENCIES

Technical Project Planning & Roadmapping · Requirements Analysis · Scope & Change Management · Risk Identification & Mitigation · Stakeholder Communication · Cross-Functional Coordination · Technical Decision Support · Product & Business Alignment · System Architecture · Reproducible Engineering · Explainability & Safety by Design · Multilingual Collaboration

TECHNICAL SKILLS

Languages	Python, Dart, C++, C#, SQL, JavaScript
AI / ML	PyTorch, EfficientNet, Computer Vision, Grad-CAM, OpenCV, Scikit-learn, Alumentations, Explainable AI
Reinforcement Learning	PPO, SAC, A2C, Multi-Agent RL, CTDE, Safe-RL, LSTM policies, Self-Play, Domain Randomization
Simulation	Discrete-Event Simulation, Entity-Component-System, Deterministic Replay, Procedural Generation, Sensor Simulation, Operations Research (Erlang C, M/M/c, Little's Law)
Backend	FastAPI, PostgreSQL, SQLAlchemy, Redis, SQLite, REST APIs, Clean Architecture, Multi-Tenant Systems, RBAC
Visualization	Three.js, Pygame, PySide6, Streamlit, Flutter, Unreal Engine 5, Firebase
Engineering	Docker, Git, Pytest, Cl, Ruff, MyPy, ONNX

KEY PROJECTS

High-Throughput Sorting-Center Digital Twin

Python · PyTorch · PPO · Multi-Agent RL · LSTM · Discrete-Event Simulation · Three.js — [repository](#)

- Built a logistics digital twin of a 400-lane sorting centre on a custom deterministic discrete-event engine with an Entity-Component-System architecture, making every run reproducible from a single seed.
- Modelled ~96,400 items/hour against a 100,000 items/hour design target, validated against classical queueing-theory baselines.
- Implemented PPO-based parcel routing, multi-agent coordination using centralised training with decentralised execution (CTDE), and an LSTM predictive-maintenance model.
- Added a fault-injection Chaos Agent, time-machine replay with scenario branching, a 2D operations centre and a browser-based 3D twin.
- 530+ automated tests documented in the repository, with Ruff and MyPy in the toolchain.

Strict Drone Safe-RL — Autonomous Control Research Platform

Python · PyTorch · PPO · LSTM · Gymnasium · ONNX · Docker — [repository](#)

- Trained drone-control policies under wind, sensor drift, control latency, motor degradation and GPS-denied navigation.
- PPO with an LSTM actor-critic under Automatic Domain Randomization, plus a Rapid Motor Adaptation-inspired module for online adaptation to changing dynamics.
- Simulated IMU, VIO and LiDAR sensing; five-stage fault-injection curriculum; TrainingGuardian stability monitoring.
- Hard safety limits and emergency fail-safe design, with an ONNX export path for edge deployment.

Simulation and research platform. No real-world autonomous flight validation is claimed.

Fracture Detection AI — Explainable Medical Imaging Product

Python · PyTorch · EfficientNet-B0 · Grad-CAM · Streamlit · SQLite · ReportLab — [repository](#)

- End-to-end computer-vision product for bone-fracture detection on the 4,083-image FracAtlas X-ray dataset.
- Measured results: 90.21% accuracy, 89.31% ROC AUC, 76.67% precision, 63.89% recall, 69.70% F1.
- Grad-CAM overlays make every prediction inspectable; shipped as a Streamlit application and Telegram bot with automated PDF reporting and local SQLite prediction history.
- Trilingual interface (English, Arabic, Russian); 21/21 documented tests passing.

Research and portfolio software. Not a certified medical device.

HMS — Multi-Tenant Enterprise ERP & AI Platform

Python · FastAPI · PostgreSQL · SQLAlchemy · Redis · React · Docker — [repository](#)

- Multi-tenant ERP covering inventory, warehousing, procurement, sales/CRM, finance, maintenance and HR in one Clean Architecture codebase.
- Enforced tenant isolation with PostgreSQL Row-Level Security at the database layer rather than trusting application code.
- Saudi ZATCA Phase 1 e-invoicing compliance, RBAC and audit logging, full Arabic/English RTL support.
- Offline rule-based advisor that runs without any external LLM, with provider-agnostic LLM agents as an optional layer.

AI Evolution Racing Lab — Explainable RL Research

Python · PyTorch · Stable-Baselines3 · PPO/SAC/A2C · Gymnasium — [repository](#)

- Self-evolving simulation where procedurally generated worlds and adaptive difficulty memory drive RL driver training across generations.
- Explainability layer combining policy-entropy confidence, input-gradient saliency and neural activation visualisation.
- AI tournaments, evolution timeline and automated research reports.

Multi-Agent Autonomous Simulation Platform

Python · PyTorch · Multi-Agent RL · PySide6 · SQLite — [repository](#)

- Multi-agent self-play across navigator, observer and interceptor roles in procedurally generated environments with dynamic weather and day/night cycles.
- Decision-probability and value-estimate logging with SQLite episodic memory, surfaced through a real-time PySide6 monitoring dashboard.

PROFESSIONAL EXPERIENCE

Full-Stack Developer / Technical Owner | HMS — medical equipment manufacturer

January 2024 — Present

- Own the corporate web product end to end: gather requirements from internal stakeholders and translate them into scoped, estimated work with realistic deadlines.
- Make the technical and architectural calls; build front end and back end, responsive layout, and integration with internal company systems.
- Carry every feature through development, testing, release and support, surfacing technical risk before it reaches a release.
- Agree priorities with stakeholders and explain technical constraints in business language.

Full-Stack Developer / Product Delivery Owner | AuraGift — digital gifting e-commerce

January 2024 — Present

- Own the client side and UX of a premium digital-gifting platform, running features from requirement through QA to production deploy.
- Built the interface and UX layer — responsive layout, animation, cross-browser behaviour and performance tuning.
- Set up and maintain hosting and deployment on Firebase Hosting, giving a predictable, repeatable release process with no downtime.

Technical Project Lead / Full-Stack Developer | HMS

January 2023 — June 2023

- Led a team of 8 on a corporate web product while contributing to the build as a full-stack developer.
- Planned phases, decomposed and assigned work, and tracked schedule, quality and milestone delivery.
- Independently managed a project budget of roughly \$10,000 — planning spend, allocating budget and paying contributors.
- Unblocked developers on hard problems and took on critical modules where extra support was needed.

Technical Project Manager / Product Engineer | Freelance · Self-employed

June 2022 — Present

- Single point of accountability between client and delivery: discovery interviews, requirements formalised into SRS, user stories and acceptance criteria, phase planning and risk management.
- Led a team of 10 on an AI logistics system with a budget of roughly \$13,000 over 11 months (January–November 2024).
- Design solution architecture and make the stack and integration decisions; build AI features — ML models in product services, LLM tooling, computer vision and explainable AI.
- Build cross-platform Flutter applications for mobile, web and desktop, plus backend services, relational databases and REST integrations.

Unreal Engine Developer / Independent Product Owner | Independent (indie)

January 2020 — Present

- Ship own games and desktop products from concept to store listing, promotion and updates.
- Built networked multiplayer with server authority (Advanced Sessions) and cloud player data (Microsoft PlayFab) — a distributed client-server system assembled solo.
- Published 3 Android titles on Google Play with 5,000+ combined downloads, handling store requirements and release management end to end.
- Authored and published an Unreal Engine (Blueprints) course on Udemy; ran Google Ads acquisition campaigns.
- Key project — PROJECT NOVA: multiplayer sci-fi shooter on Unreal Engine 5 (C++ and Blueprints) with character replication, matchmaking, Nanite/Lumen and Niagara effects.

Media Team Lead | Resala Charity Organization, Egypt

January 2020 — December 2020

- Led the media function at one of the largest charities in Egypt and the Middle East: organised team workflow, assigned ownership and tracked delivery.
- Coordinated people and activities across campaigns and events, managing deadlines under limited resources.
- Communicated with organisational stakeholders and resolved conflicts within the team.
- Completed a six-month leadership and management development programme.

EDUCATION

Bachelor — Computer Science & Engineering

Tomsk State University of Control Systems and Radioelectronics (TUSUR) · 2026

Relevant coursework: Algorithms, Data Structures, Operating Systems, Software Engineering, Artificial Intelligence, Computer Vision, Machine Learning, Networking, Databases.

CERTIFICATIONS & RECOGNITION

- IBM Professional Course — Certificate of Appreciation
- Resala Charity Organization — Certificate of Appreciation, Media Development Manager
- Tomsk State University (TUSUR) — Recognition for contributing to an educational Virtual Reality (VR) system

LANGUAGES

Arabic — Native • **English** — C2, full professional • **Russian** — B2, upper-intermediate